Mock Final Exam #3 – Machine Learning (MDS + BDMA + MIRI)

Duration: 1h30m

All questions have equal weight. Justify all your answers. Be concise and clear.

Question 1

A dataset contains highly correlated features. You decide to train the following models:

(a) Ridge regression

(b) Lasso regression

(c) Decision tree

- How would each model behave with such features?
- Which model would likely generalize best and why?
- Propose any preprocessing steps you would apply to improve model behavior.

Question 2

You are using Bayesian learning to model data with unknown parameters θ.

(a) Define the posterior distribution and the predictive distribution.

(b) Why is the predictive distribution often harder to compute?

(c) Explain in practical terms what each represents in model deployment.

Question 3

You train a neural network and a k-NN model on the same classification dataset. On the test set: - k-NN achieves 98% accuracy - Neural network achieves 85% accuracy

- Provide three plausible explanations for this performance gap.
- How would you test which explanation is correct?
- What would you try to improve the neural network's performance?

Question 4

For each of the following modifications, explain its effect on bias and variance. Assume models are otherwise well-optimized.

(a) Increasing the number of trees in a Random Forest

(b) Increasing the number of layers in an MLP

(c) Using a smaller k in k-NN

(d) Switching from L2 to L1 regularization in logistic regression

Question 5

Mark each statement as **True** or **False**. Justify each answer briefly.

(a) A decision boundary created by a decision tree is always axis-aligned.

(b) k-fold cross-validation provides an unbiased estimate of the true generalization error.

(c) Bagging reduces both bias and variance.

(d) An SVM with a polynomial kernel can fit any training set with zero error if the degree is high enough.

(e) A higher training error always implies a higher test error.

(f) In LOOCV, the validation error is always lower than the training error.